

# YUANJUN (JOHNNY) DAI

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## EDUCATION

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|---------------------------------------------------------------------|----------------------------|
| Ph.D. Candidate in Computer Science (AI Infrastructure)             | 08/2019 – Dec. 2026 (exp.) |
| Case Western Reserve University                                     |                            |
| M.S. in Electrical and Computer Engineering (Computer Architecture) |                            |
| University of Pittsburgh                                            |                            |
| B.S. in Electrical Engineering                                      |                            |
| Northeastern University                                             |                            |

## SKILLS

|                       |                                                                                         |
|-----------------------|-----------------------------------------------------------------------------------------|
| Languages             | Python, C/C++                                                                           |
| Agentic AI            | LangGraph, Multi-Agent Orchestration, Multi-Model Routing, Prompt Engineering           |
| ML Frameworks         | PyTorch (DDP/FSDP), vLLM, Megatron-LM, NCCL, BytePS, scikit-learn                       |
| Parallelism Paradigms | Data Parallel (DDP/FSDP), Tensor Parallel (TP), Pipeline Parallel (PP), Hybrid Parallel |
| Infrastructure        | Kubernetes, Docker, CI/CD Pipelines (GitHub Actions), ArgoCD, Slurm                     |
| Systems & APIs        | FastAPI, REST API, gRPC                                                                 |
| Observability         | eBPF (Kernel Tracing), Prometheus, Grafana, DCGM Exporter, Intel RAPL                   |
| Cloud & HPC           | Lambda Labs HPC, Ohio Supercomputer Center, NSF FABRIC, CloudLab                        |
| Networking            | SDN (Software-Defined Networking), P4, NPU/ASIC (Broadcom BCM SDK)                      |

## WORK EXPERIENCE

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| Software Engineer II                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 06/2015 – 08/2016          |
| Cisco Systems, Inc.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                            |
| <ul style="list-style-type: none"><li>Developed NPU/ASIC features for a 100 Gbps core router (Azure, AWS, Alibaba Cloud) using Broadcom BCM SDK.</li><li>Contributed to QoS and ECMP modules, reducing forwarding latency under heavy load.</li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                            |
| Software Engineer Intern                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 10/2014 – 05/2015          |
| Broadcom (Emulex)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                            |
| <ul style="list-style-type: none"><li>Re-architected RTOS data acquisition with DMA + circular-buffer, <b>reducing CPU load by 30%</b>; ported BladeEngine ASIC management CLI from Linux to Windows Server.</li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                            |
| Technical Product Manager                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 07/2018 – 01/2019          |
| Xiaomi                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                            |
| <ul style="list-style-type: none"><li>Defined IoT appliance product requirements via competitive analysis and user research; assessed technical feasibility and delivered architecture proposals to align engineering and business stakeholders.</li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                            |
| Product Manager                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 09/2016 – 06/2018          |
| Midea Group                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                            |
| <ul style="list-style-type: none"><li>Led technical PM for <b>Fun Oven II</b> across ML model training, Docker deployment on Alibaba Cloud, mobile app, and culinary algorithm design; won <b>Red Dot Design Award</b>, debuted at 2018 AWE Show · <a href="#">Red Dot ↗</a> · <a href="#">PR Newswire ↗</a></li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                            |
| Research Assistant, AI Infrastructure & ML Systems                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 08/2019 – Dec. 2026 (exp.) |
| Case Western Reserve University                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                            |
| Research focus: <b>security, optimization, and observability</b> of <b>ML/LLM</b> training and inference infrastructure.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                            |
| <b>DYNAMIX – Reinforcement Learning-Based Adaptive Batch Scheduling for Distributed ML</b>   <a href="#">Paper</a> / <a href="#">Code</a>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                            |
| <ul style="list-style-type: none"><li>Designed a <b>PPO-based Reinforcement Learning (RL)</b> scheduler monitoring hardware signals (CPU/memory, bandwidth, retransmission rate via <b>eBPF</b> probes) and statistical signals (loss convergence, generalization) to dynamically adapt batch size</li><li>Co-scales learning rate via <b>Linear Scaling Rule</b>; evaluated on <b>PyTorch DDP</b> and <b>Parameter Server (BytePS)</b></li><li>Achieves <b>46% wall-clock speedup</b> and <b>~6% accuracy gain</b> across <b>8–64 A100 GPUs</b> over fixed batch size baseline</li></ul>                                                                                                                                                                                                                |                            |
| <b>AI-Powered Job Hunting Agent</b>   <a href="#">Code</a> / <a href="#">Report</a>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                            |
| <ul style="list-style-type: none"><li>Built a <b>multi-agent AI platform</b> for automated job discovery, semantic matching, and resume optimization across 5 platforms</li><li>Designed a <b>LangGraph-based orchestration pipeline</b> with specialized agents for collection, scoring, and refinement</li><li>Developed a <b>6-dimension LLM-based job matching framework</b> via careful <b>prompt engineering</b> evaluating skills, experience, seniority, authorization, and company preference</li><li>Implemented a <b>capability-aware multi-model routing pipeline</b> for analytical planning and recruiter-oriented resume optimization</li><li>Engineered <b>FastAPI</b> backend with cross-platform deduplication, health monitoring endpoints, and deployment-ready automation</li></ul> |                            |

## Domain-Adapted LLMs for Clinical Risk Adjustment Coding | [Paper](#)

- Built a 115-label HCC multi-label classifier on highly imbalanced EHR data (MIMIC-V), improving long-tail performance via **inverse-frequency weighted binary cross-entropy loss** and **per-label threshold calibration**, achieving **0.775 macro-F1 / 0.742 micro-F1 / 0.9362 macro-AUC**.
- Trained **PubMedBERT** using **PyTorch DDP** on 2×NVIDIA RTX A6000 GPUs for efficient scaling, while fine-tuning **Qwen-7B** using distribution-aware **few-shot prompting** to **mitigate hallucination** and enforce structured HCC outputs with improved sensitivity to sparse labels; scaled full-parameter training via **NeMo + Megatron-LM** (TP=1, PP=1) with **PyTorch FSDP (ZeRO-3)**, **FlashAttention**, and **activation checkpointing**, enabling training under GPU memory constraints (avoiding OOM).

## Scalable LLM Serving Infrastructure on Kubernetes | [Code](#) / [Video](#)

- Independently designed a self-hosted **K8s** LLM inference platform (NGINX → **FastAPI** → **vLLM**) with **GitOps CI/CD** (**ArgoCD** + GitHub Actions) across 50+ deployments; validated on **RTX 4050** (**GPU time-slicing**, **Horizontal Pod Autoscaler (HPA)** 1↔2) and **Lambda Labs A100** (**Multi-Instance GPU (MIG)** 7×1g.5gb, **HPA** 1↔7) with **91.3% peak GPU compute** under extreme stress
- Built 4-layer GPU observability (**DCGM Exporter** + **Prometheus** + **Grafana**: GPU hardware, vLLM metrics, FastAPI gateway, K8s cluster)

## PRISM – Priority-Based Selective Reliability for Distributed ML Training | [Paper](#)

- Designed a transport-layer library (**NCCL/Gloo** → **UDP**) hooking into **PyTorch DDP**'s AllReduce at bucket-ready events; classifies gradients by magnitude into high-priority (full **SACK**-based retransmission) and low-priority (local compensation via cached values)
- Supports kernel-bypass (**LibVMA**) and kernel-based (**GSO/GRO**) variants; achieves **8–15% throughput gain for CNNs** and **15–21% for LLMs** over TCP, matching NDP latency at 144-worker scale

## In-Kernel eBPF Sketch Systems for Network and System Resource Monitoring | [Paper \(PSketch\)](#) / [Code](#) · [Paper \(eHashPipe\)](#) / [Code](#)

- **PSketch**: first in-kernel priority-aware sketch using **eBPF**; high-priority flows tracked via in-kernel hashmap ( $O(1)$ ), low-priority via **3-layer Count-Min Sketch**; enables  **$O(n)$  Top-K detection** with sub-linear memory; achieves **96% Top-K accuracy**, **96.4% retransmission recall**, <1.1% throughput overhead at 10 Gbps
- **eHashPipe**: dual-pipeline **eBPF** framework with  **$O(n)$  Top-K tracking** — (1) HashPipe-based memory profiler with cascading slot eviction and ptr2size deallocation tracking, (2) per-thread CPU tracker via sched\_switch tracepoint; delivers **10–14× finer temporal resolution** than top/htop, ~11ms latency

## Training-Time Side-Channel Attacks on Distributed ML Systems | [Paper](#) / [Code](#)

- **Analyzed DML computation/communication behavior** in **MXNet BSP** Parameter Server: instrumented push/pull primitives to profile per-layer network traffic patterns; modified MXNet source code for ground-truth collection to capture **DML network behavior** at layer granularity
- Profiled **ML energy consumption** via **Intel RAPL** (PP0+DRAM, 50μs via powercap/sysfs); applied **sliding window** algorithms to isolate forward passes; used **band-pass filtering** and **K-means clustering** for noise reduction and layer segmentation; applied **polynomial regression** for per-layer and **activation function energy profiling**; classified layer types via **Decision Tree** and recovered Conv/FC dimensions via weighted-average **MLP ensemble** — exposing DNN architecture stealing surfaces in MLaaS environments

## PUBLICATIONS

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8 papers total, 7 first-author, 5 published at IEEE CCNC, IEEE ICNC, ACM TOIT, and Springer SecureComm; 3 under review (incl. ACM TOPS). Full list: [johnny-dai-git.github.io/portfolio](https://johnny-dai-git.github.io/portfolio)

## PATENTS

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Oven Temperature Control Method, Device and Computer-Readable Storage Medium — Granted CN 108594898 B | Issued July 23, 2021 | Assignee: Midea Group Co., Ltd. | Dai Yuanjun et al.